

Project Proposal

Team Name: CSE4104-7C-T03

Project Title

AI Study Material Generator

Problem Statement

Students often struggle to prepare organized and effective study materials from large lecture notes, PDFs, slides, and textbooks. Preparing summaries, important questions, quizzes, and revision notes manually takes a significant amount of time and effort.

Existing study platforms mainly provide static resources and lack intelligent automation and personalized learning support. Most systems cannot automatically analyze uploaded educational content and generate customized study materials for students.

Artificial Intelligence is highly relevant for solving this problem because AI can understand educational content using Natural Language Processing (NLP), summarize information, generate MCQs, create flashcards, and provide smart study recommendations automatically.

The proposed system aims to simplify the learning process by generating intelligent study materials quickly and efficiently.

Objectives

The main objectives of the project are:

- Build an AI-powered study material generation system
- Generate automatic summaries from notes and PDFs
- Create AI-generated MCQs and quiz questions
- Generate flashcards and short notes
- Provide personalized study assistance
- Improve student learning efficiency and productivity

Proposed Features

The proposed system will include the following features:

- User authentication system
 - Upload PDF, lecture notes, and documents
 - AI-generated summaries
 - Automatic MCQ generator
 - Quiz generation system
 - Flashcard generator
 - Important question prediction
 - AI chatbot for student assistance
 - Dashboard for managing study materials
 - Download generated study resources
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Technology Stack

Frontend

- HTML
- CSS
- JavaScript
- React.js

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JWT

AI Integration

- OpenAI API

Deployment

- Frontend: Vercel
- Backend: Render
- Database: MongoDB Atlas

Version Control

- Git & GitHub

Expected Outcome

The expected outcome of this project is to create an intelligent educational assistant that helps students prepare study materials automatically and efficiently.

Users will benefit by:

- Saving study preparation time
- Receiving personalized learning support
- Improving revision efficiency
- Accessing smart AI-generated educational content
- Practicing through automatically generated quizzes and MCQs

The system will improve traditional study workflows by reducing manual effort and providing intelligent learning assistance through AI integration.
